Comparative Analysis of CNN and Vision Transformer Architectures for Image Classification

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1 Introduction

Image classification is one of the fundamental tasks in computer vision. Its objective is to assign an input image to one of several predefined categories. This task is important because it serves as the basis for many real-world applications, including medical image analysis, industrial quality inspection, autonomous driving, retail analytics, and security systems. In the Hugging Face computer vision course, image classification is presented as one of the core computer vision tasks, while images themselves are described as spatially structured data that require specialized methods for effective processing. [1]

The growing complexity of image-based applications has led to the development of two dominant families of architectures for classification: **Convolutional Neural Networks (CNNs)** and **Vision Transformers (ViTs)**. CNNs have long been the standard approach because they are naturally suited to spatial data and can efficiently learn local visual patterns. More recently, Vision Transformers have emerged as a strong alternative by adapting the Transformer architecture to visual tasks and modeling long-range dependencies more directly [2].

The purpose of this report is to compare CNN and ViT architectures from both a theoretical and practical perspective in order to determine which approach is more suitable for image classification under constrained computational resources but high accuracy requirements.

2 Theoretical Background

2.1 Convolutional Neural Network

CNNs are specifically designed for image-like data. Since images are spatially organized arrays of pixels, CNNs exploit this structure by focusing on local neighborhoods and gradually building more abstract feature representations. The Hugging Face course emphasizes that image data contain rich spatial information and that CNNs are the traditional deep learning models used for image tasks. [1]

The operation of CNNs is based on three core ideas: **convolution**, **pooling**, and **hierarchical feature learning**.

Convolution is the main mechanism for feature extraction. A convolutional filter slides across the input image and detects local patterns such as edges, textures, and shapes. As more convolutional layers are stacked, the model learns increasingly complex visual structures.

Pooling reduces the spatial size of feature maps and helps retain the most important information while discarding minor variations. Pooling also makes the model less sensitive to small shifts or distortions in the input. This contributes to the relative translation robustness of CNN-based systems.

Hierarchical feature learning means that earlier layers detect simple patterns, whereas deeper layers combine them into more meaningful object-level representations. This hierarchical organization reflects the compositional nature of images.

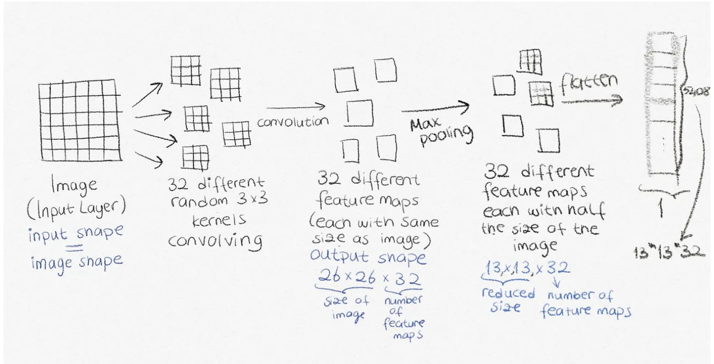

Figure 1: CNN[1]

Representative CNN architectures include **ResNet** and **EfficientNet**. ResNet introduced residual learning to make very deep networks easier to optimize [3]. EfficientNet later improved efficiency by scaling depth, width, and resolution in a more principled manner [4].

2.2 Vision Transformers

Vision Transformers adapt the Transformer architecture, originally developed for sequence modeling, to image analysis. Instead of processing an image through local filters, ViT divides the image into a sequence of fixed-size patches and treats these patches as input tokens [2].

The course explains that transformer-based vision models convert images into sequence-like representations and enrich them with positional information before processing them through Transformer blocks.

The main building blocks of ViT are the following:

Patch embedding. The input image is divided into small patches, and each patch is projected into a vector representation. This converts the 2D image into a sequence of visual tokens. The course describes patch-wise embedding as the step that transforms images into tokenized inputs for the Transformer.

Positional embeddings. Because Transformers do not inherently encode spatial order, positional information must be added to indicate where each patch came from. The course refers to row and column embeddings as a way to preserve this spatial structure.

Self-attention. Self-attention allows each patch token to interact with every other patch token. This is the fundamental reason why ViTs are strong at modeling long-range and global relationships in the image. Unlike CNNs, which build global understanding gradually, ViTs can directly compare information from distant regions.

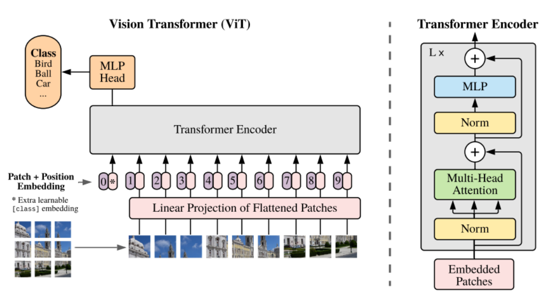

Figure 2: ViT [1]

Well-known architectures in this family include **ViT**, **DeiT**, and **Swin Transformer**. DeiT was proposed to make Vision Transformers more data-efficient through improved training procedures and knowledge distillation [5]. Swin Transformer introduced shifted window attention to improve computational scalability while preserving the advantages of hierarchical Transformer representations [6]. The course likewise describes Swin as a more computationally efficient variant that applies self-attention within local windows and uses shifted windows to enable broader contextual interaction.

3 Comparative Analysis

3.1 How each architecture "sees" an image

A CNN sees an image as a continuous spatial structure. It starts from local pixel neighborhoods and progressively constructs higher-level feature maps. This makes CNNs especially well suited to detecting local structures such as contours, corners, and textures. Their behavior reflects the idea that visual meaning often emerges from relationships among nearby pixels.

A ViT, by contrast, first converts the image into a sequence of patches. Once patch embeddings and positional information are created, the model operates on tokens rather than directly on the 2D image. Through self-attention, it can compare and integrate information from distant image regions in a single layer. Therefore, ViT is less constrained by locality and more naturally captures global context [2].

In summary, CNNs are inherently local-first models, whereas ViTs are global-context models.

3.2 Transfer learning and fine-tuning

Transfer learning is essential for both CNNs and ViTs in practical applications. Rather than training models from scratch, researchers commonly start from pretrained weights and adapt the model to a target dataset.

For CNNs, transfer learning is usually stable and efficient because early convolutional layers tend to learn generic visual features such as edges, gradients, and textures. These features are often transferable across tasks. In many cases, only the final classification layers need to be retrained, while the earlier layers can be partially or fully frozen.

For ViTs, transfer learning is also effective, especially when a pretrained model has already learned broad visual representations from a large dataset. The Hugging Face course explains that pretrained Vision Transformers can be fine-tuned by freezing most of the model or by updating the full network with a smaller learning rate when the target domain differs substantially from the pretraining domain. [1]

However, ViTs are often more sensitive to fine-tuning strategy. Their performance depends more strongly on hyperparameter selection, regularization, and augmentation. This aligns with DeiT, which specifically addressed data-efficient training and better optimization of transformer-based image models [5].

Therefore, while both CNNs and ViTs benefit from transfer learning, CNNs are generally considered more forgiving during fine-tuning, whereas ViTs may require more careful optimization choices.

3.3 Data requirements

When trained from scratch, Vision Transformers generally require more data than CNNs. This difference can be explained by inductive bias.

CNNs encode assumptions that are very appropriate for images, such as locality and translational equivariance. These assumptions act as useful constraints and allow CNNs to learn efficiently even when the training set is not extremely large. The course directly states that CNNs benefit from these inductive biases, whereas ViTs do not. As a result, CNNs tend to outperform ViTs below a certain data threshold. [1]

ViTs, on the other hand, are more flexible and scalable, but this flexibility comes at a cost. Because they lack the same built-in assumptions about image structure, they need more data to learn robust visual representations. Large-scale pretraining helps compensate for this weakness, which is why modern ViTs often perform very well in practice when pretrained on large image corpora [2].

DeiT is a particularly important example because it showed that Vision Transformers can become competitive even on ImageNet-1k when trained with better regularization, augmentation, and distillation strategies [5].

Thus, the general conclusion is clear: **for training from scratch, ViTs usually need more data; with strong pretraining, the gap becomes much smaller**.

3.4 Computational efficiency

CNNs are often more practical for deployment because convolution is relatively efficient and exploits the structure of image grids well. The Hugging Face course also highlights that image size, model size, memory constraints, and compute requirements are important practical concerns in computer vision pipelines. It stresses that larger images and larger models increase memory use and training cost, which is especially problematic in resource-constrained environments. [1]

Classical Vision Transformers are often computationally expensive because self-attention is applied across all image patches, which increases complexity significantly as image size grows. This is one of the reasons why early ViTs were difficult to use on smaller hardware.

Swin Transformer addresses this issue by restricting self-attention to local windows and then shifting the windows between layers to recover cross-window interactions. This reduces computational cost while preserving much of the representational power of Transformers [6]. The course similarly describes Swin as a more efficient Transformer variant that uses local window attention and shifted windows for broader contextual learning [1].

3.5 Robustness to augmentation

Data augmentation is a major factor in image classification performance. The Hugging Face course explicitly states that augmentation is crucial for improving the performance and generalization of CNNs in image classification. It also lists common image augmentation techniques such as flipping, cropping, random rotation, brightness changes, shifts, and zoom.

For CNNs, augmentation is especially important because it increases robustness to common visual transformations while preserving the model's useful local biases. Operations such as horizontal flips, random crops, and small rotations help CNNs generalize better to real-world visual variation. The course presents augmentation as a cornerstone technique for practical CNN-based classification systems. [1]

For ViTs, augmentation is arguably just as important, and in many cases even more important. Since ViTs do not have the same strong spatial inductive biases as CNNs, they rely more heavily on sufficiently diverse training signals. This is consistent with the DeiT paper, where data-efficient training was made possible through carefully designed augmentation and regularization strategies [5].

4 Practical Recommendations

In experiment on CIFAR-10 using pretrained models, the results were as follows:

1. ResNet18 - 11,181,642 params
2. EfficientNet-B0 - 4,020,358
3. ViT-Tiny - 5,526,346
4. DeiT-Tiny - 5,526,346
5. Swin-Tiny - 27,527,044 (Not trained due to computational cost and time)

Model	Accuracy	F1-score	Parameters	Time/epoch (s)
EfficientNet-B0	0.9604	0.9604	4,020,358	4056.03
ViT-Tiny	0.9506	0.9507	5,526,346	2539.74
DeiT-Tiny	0.9440	0.9440	5,526,346	2631.24
ResNet18	0.9336	0.9334	11,181,642	217.16

Table 1: Comparison of CNN and Transformer-based models on CIFAR-10

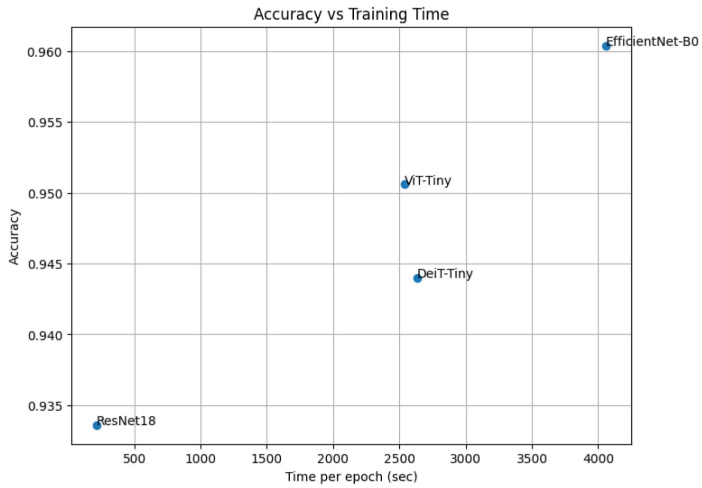

Figure 3: Accuracy and training time of each model

If the main priority is maximum accuracy, then **EfficientNet-B0** is the most justified choice in this setting. It achieved the best performance among all tested models.

If the goal is to maintain high performance while adopting a Transformer-based approach, **ViT-Tiny** is a strong option. It provided better accuracy than ResNet18 and DeiT-Tiny, although it still did not surpass EfficientNet-B0 in this experiment.

If the priority is **speed and lower training cost**, then **ResNet18** remains a practical and reliable baseline. It is not the most accurate model, but it is computationally efficient and comparatively easy to fine-tune.

This leads to an important conclusion: **ViT is not always better than CNN**. Transformer-based models are powerful, especially when supported by large-scale pretraining and carefully selected training strategies, but CNNs remain highly competitive, especially on medium-sized datasets and under limited computational budgets. Your experimental results confirm this directly, since the best model in the comparison was a CNN rather than a ViT.

Figure 4: ResNet18 before and after training

Figure 5: EfficientB0 before and after training

5 Conclusion

The overall conclusion is that there is no universally best architecture. The optimal choice depends on the size of the dataset, the availability of pretrained models, the computational budget, and the target trade-off between speed and accuracy. Under medium-scale data and limited resources, CNNs remain highly competitive and may even outperform Vision Transformers in practice. Under larger-scale settings with strong pretraining, ViTs become increasingly attractive.

References

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